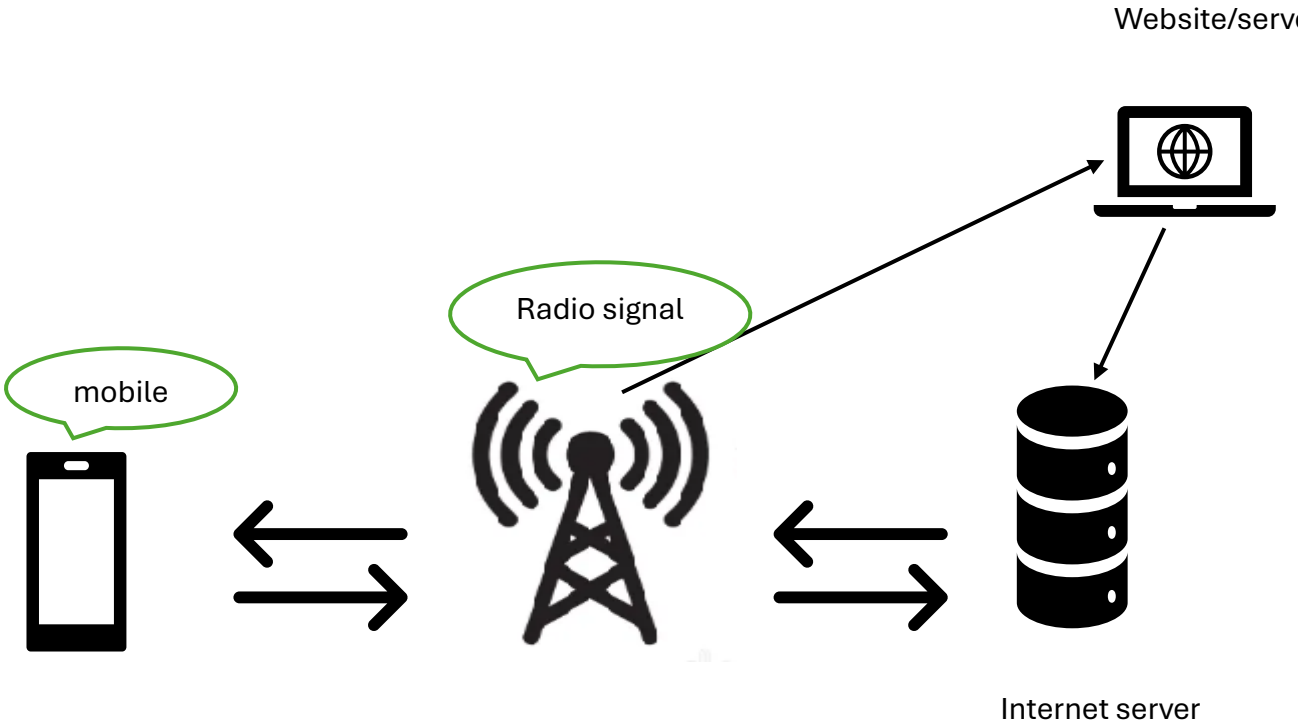
1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



- Mobile data is turned on.
- Phone send signal
- Signal reaches nearest mobile tower
- Tower connects to internet servers
- Data is sent back to the phone

2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Device	Network Type	Description
Smartphone	WAN and LAN	Connects to cellular towers over long distances (WAN) and links to home or office Wi-Fi (LAN).
Smart TV	LAN	Connects to your home Wi-Fi or Ethernet cable to stream content from local devices or routers.
Wi-Fi Router	WAN and LAN	Bridges your local home devices together (LAN) and connects your entire home to the broader internet (WAN).
Laptop	LAN	Connects to local Wi-Fi networks or office networks to share files, print, and access the internet gateway.
Smartwatch	PAN	Connects directly to your smartphone via Bluetooth to sync data over short distances centered around a single person.

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Feature	Internet	Intranet
Definition	A global network of linked computers.	A private network inside an organization.
Access	Public; open to anyone.	Private; restricted to authorized users.
Users	Unlimited, billions of users worldwide.	Limited to a specific group or organization.
Security	Less secure; higher risk of viruses and hackers.	Highly secure; protected by firewalls.
Traffic	Very high traffic.	Low traffic.
Example	Google or YouTube	College student portal or internal grade system

College Network Example:

Internet Usage: When you sit in the college library and use public WiFi to browse youtube, checking for weather updates or search on google you are using Internet.

Intranet Usage: When you log into the college portal using your student ID to check your exam grades, view class schedules or accessing internal library book records you are using the campus intranet. This portal does not work for people outside the college network.

4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

When you search for food on Zomato, your phone translates your types words into digital request

And sends it over the air via Wi-Fi or cellular networks to your internet provider.

This provider routes the request through underground internet cables directly to Zomato's central servers.

The servers process your request, search their database by matching restaurants and bundle the results together.

Finally, this data travels back through the same network cables and wireless signals to your device, where your app instantly displays the restaurant list on your screen.